🚀 AWS DataSync - Overview

📌 What is AWS DataSync?

AWS DataSync is a fully managed data transfer service that **automates and accelerates** moving data between **on-premises storage** and **AWS services** (like Amazon S3, EFS, FSx, and more). It is designed for **large-scale, secure, and efficient** transfers, both **one-time** and **recurring/scheduled**.

🔹 Key Features

Feature	Description
✅ **Fast Transfers**	Up to 10× faster than standard tools like rsync or SCP
🔁 **Automated Scheduling**	Recurring transfer jobs (hourly, daily, etc.)
🛡️ **Encryption**	Data is encrypted in transit via TLS
📦 **Built-in Compression**	Reduces bandwidth usage during transfer
📋 **Metadata & ACL Transfer**	Preserves timestamps, permissions, ownership, etc.
🔄 **Incremental Sync**	Only copies changed data after the first sync
🧠 **Monitoring & Logging**	Integrated with **Amazon CloudWatch**

📍 Supported Data Sources & Destinations

Source / Destination	Services
On-Premises Storage	NFS, SMB (file shares)
AWS Services	S3, EFS, FSx for Windows, FSx for Lustre

Between AWS Regions or Accounts	Yes, cross-region and cross-account supported

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🧩 How AWS DataSync Works

1. **Deploy an agent**:
   - For **on-premises**: Install the **DataSync agent VM** (VMware, Hyper-V, KVM).
   - For **AWS-to-AWS**: No agent needed.
2. **Create a task**:
   - Specify **source** and **destination** locations.
3. **Configure task settings**:
   - Choose filters, schedule, bandwidth limits, etc.
4. **Start the task**:
   - Monitor progress in the console or via CloudWatch.

🔄 Common Use Cases

Use Case	Description
Data Migration	Move existing datasets to AWS (e.g., file servers to S3)
Backup	Periodically sync on-prem files to S3 or EFS
Hybrid Architectures	Keep data in sync between AWS and on-prem systems
Analytics/Data Lake Ingestion	Feed files into S3 for Athena, Redshift, or Glue
Cross-region Replication	Move data between AWS regions/accounts

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🔒 Security Features

- **TLS 1.2** encryption in transit
- **No data written to disk** on agent
- Supports **IAM** roles and fine-grained permissions
- VPC endpoints for private networking
- CloudTrail logging for auditing

💰 Pricing Overview (2025)

Component	Cost
Data Transfer	**$0.0125 per GB** (in most regions)
Free Tier	No dedicated free tier (charges apply from first GB)
Additional Costs	Standard storage costs (e.g., S3), and data egress if transferring out

🧾 [Official Pricing Page]

🔄 AWS DataSync vs Other AWS Services

Service	Best For	Notes
DataSync	Large-scale automated transfers	File-based (NFS/SMB)
AWS DMS	Database migrations	Structured data (tables/SQL)
AWS Transfer Family	FTP/SFTP integrations	Real-time client-based access
AWS Snow Family	Offline data transfers	Use when network bandwidth is limited

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🧪 Quick Start Example: On-Prem NFS to Amazon S3

1. **Deploy the DataSync agent** on your local VM.
2. **Activate** the agent in AWS Console using its activation key.
3. **Create source location** (e.g., on-premises NFS).
4. **Create destination location** (e.g., Amazon S3 bucket).
5. **Create a task** and configure transfer settings.
6. **Run the task** (once or schedule it).
7. **Monitor** with **CloudWatch** and **DataSync logs**.

📝 Key Notes for Study & Interviews

- AWS DataSync = **file-based data movement tool** (not DB-specific).
- Designed for **automated**, **reliable**, and **scalable** transfers.
- You need an **agent** for on-premises sources.
- **Faster and more robust** than rsync, SCP, FTP, etc.
- DataSync **preserves metadata** and permissions.

Sure! Here's a **Step-by-Step Migration Guide using AWS DataSync** for moving files from an **on-premises file server (NFS or SMB)** to **Amazon S3, EFS, or FSx**.

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\# 🛠️ **Step-by-Step Migration Guide: AWS DataSync**

📍 Example Scenario:

Migrate data from an **on-premises NFS file server** to **Amazon S3** using **AWS DataSync agent**.

✅ **STEP 1: Prepare Your Environment**

On-Premises:

- A file server (NFS or SMB) with data you want to migrate.

- A virtual machine platform (VMware, Hyper-V, or KVM) to run the **DataSync Agent**.

In AWS:

- An **S3 bucket**, **EFS file system**, or **FSx** as your destination.

- Proper **IAM permissions** and **VPC access**.

📦 **STEP 2: Deploy the AWS DataSync Agent**

Option 1: For On-Premises

1. **Download the DataSync Agent VM** from the AWS Console.

   - Go to **AWS DataSync > Agents > Create agent**.

2. Deploy the VM on your hypervisor (VMware/Hyper-V).

3. Boot the VM and copy the **activation URL** shown on its console.

Option 2: For AWS-AWS Transfers

- No agent is needed if **both source and destination are AWS services**.

🔗 **STEP 3: Activate the Agent**

1. In the AWS Console, go to **DataSync > Agents > Create agent**.
2. Paste the activation URL from the agent.
3. Select the **VPC and subnet** for communication with AWS.
4. Wait for the agent to become **active**.

📍 **STEP 4: Create Source Location**

1. Go to **DataSync > Locations > Create location**.
2. Choose:
   - Location type: **NFS** (or SMB)
   - Enter:
     - Server IP / hostname
     - Mount path (e.g., `/mnt/data`)

  - Select the **agent** deployed earlier.

3. Click **Create location**.

☁️ **STEP 5: Create Destination Location**

1. Go to **DataSync > Locations > Create location**.

2. Choose destination type:

  - **Amazon S3**, **EFS**, or **FSx**

3. Provide required details:

  - For S3: Bucket name, folder prefix

  - IAM role (use default or custom with `s3:PutObject`, etc.)

4. Click **Create location**.

🔁 **STEP 6: Create a Transfer Task**

1. Go to **Tasks > Create task**

2. Configure:

  - **Source** and **destination** locations

  - **Task settings**:

    - **Transfer mode**: One-time or scheduled

- **Overwrite files**, **preserve metadata**, etc.

- **Include/Exclude filters** (optional)

3. Review and **Create task**

▶️ **STEP 7: Start the Task**

1. In the console, go to **Tasks**

2. Select your task and click **Start**

- Choose: **Start with a full data transfer**

3. Monitor progress:

- Use **task execution logs**

- **CloudWatch metrics** for performance, errors

🔍 **STEP 8: Verify and Monitor**

- Ensure data appears in the destination (e.g., S3 bucket).

- Check logs for errors or skipped files.

- If needed, set the task to run on a **schedule** (daily, hourly, etc.)

🧹 **STEP 9: Clean Up (Optional)**

- Stop and delete the DataSync task and agent when done.
- Decommission the DataSync agent VM if no longer needed.
- Remove IAM roles/policies if custom roles were used.

📝 Bonus: Sample IAM Policy for S3 Destination

```json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:PutObjectAcl",
        "s3:ListBucket",
        "s3:GetBucketLocation"
      ],
      "Resource": [
```

```
        "arn:aws:s3:::your-bucket-name",
        "arn:aws:s3:::your-bucket-name/*"
      ]
    }
  ]
}
```

📘 Summary Checklist

Step	Action
✅ 1	Prepare your source & destination
✅ 2	Deploy and activate the agent
✅ 3	Create source location (e.g., NFS/SMB)
✅ 4	Create destination (e.g., S3)
✅ 5	Configure and create task
✅ 6	Run and monitor the task
✅ 7	Validate the transfer